

Laurea in Fisica ■ Bachelor's Degree Thesis

Non-reciprocal Effects in 2D Superconductors Studied by Conformal Mapping

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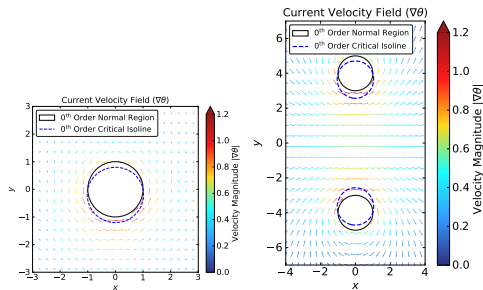
A.Y. 2025/2026

Theory & Analytical Tools (Ch. 1 & 2)

- **Ch. 1:** Ginzburg-Landau phase field $\theta(\mathbf{r})$, topological vortices, and critical velocity threshold $v_c = 1$.
- **Ch. 2:** Complex potential $f(z)$, complex velocity $W(z)$, and conformal mapping principles.

Core Thesis Findings (Ch. 3 & 4)

- **Ch. 3:** Single Pinned Vortex: breakdown of 0th-order self-consistency, 1st-order geometric feedback, and strict transport reciprocity.
- **Ch. 4:** Vortex-Antivortex Dipole: central channel collapse, directional critical current gap ($j_{c,+} \neq |j_{c,-}|$), and the Superconducting Diode Effect.



Physical system: isolated vortex and vortex-antivortex dipole under uniform transport current.

Chapter 1

Theory

Ginzburg-Landau & Topological Vortices



GL Phase Field & XY Model Limit

Below T_c , Cooper pair density freezes in the bulk ($\rho \rightarrow \rho_0$). Supercurrent is governed purely by the phase field $\theta(\mathbf{r})$, equivalent to the continuum limit of the classical XY model:

$$\psi(\mathbf{r}) = \sqrt{\rho_0} e^{i\theta(\mathbf{r})} \implies \mathbf{v} = \nabla \theta$$

Minimizing the kinetic energy $E = \frac{\rho_s}{2} \int d^2r (\nabla \theta)^2$ yields Laplace's equation:

$$\nabla^2 \theta = 0$$

Topological Defects & Critical Velocity

- **Singular Winding:** Phase changes by $2\pi n$ around defects: $\oint \nabla \theta \cdot d\mathbf{l} = 2\pi n$.
- **Core Formation:** Velocity $v = n/r$ diverges at $r \rightarrow 0$.
- **Critical Threshold:** When v reaches v_c , superconductivity breaks down and a normal region forms.
- *Dimensionless units: core radius $r_0 = 1$, critical velocity $v_c = 1$.*

Discrete Spin Lattice

The system can be modeled by a 2D lattice of classical spins, where the angle θ_i represents the phase of the order parameter. The Hamiltonian with ferromagnetic coupling $J > 0$ is:

$$H = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j)$$

Continuum Limit

- For small phase variations ($\Delta\theta_{ij} \ll 1$), we expand the cosine to quadratic order.
- In the continuum limit ($a \rightarrow 0$), the discrete sum becomes an integral over the phase gradient.
- Identifying J with the superfluid stiffness ρ_s , we exactly recover the macroscopic kinetic energy [4]:

$$E(\theta) = \frac{\rho_s}{2} \int d^2\mathbf{r} (\nabla\theta)^2$$

Chapter 2

Methods

Complex Potential & Conformal Mapping



Conjugate Stream Function $\psi(x, y)$

Since $\nabla^2 \theta = 0$, we introduce the conjugate stream function $\psi(x, y)$ to define an analytic complex potential:

$$f(z) = \psi(x, y) + i\theta(x, y), \quad z = x + iy$$

- $\psi = \text{const}$: superfluid streamlines ($\mathbf{J}_\perp = 0$)
- $\theta = \text{const}$: equipotential phase lines
- Complex velocity:

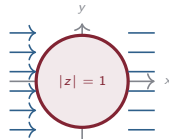
$$W(z) = \frac{df}{dz} = v_y + iv_x \implies |\mathbf{v}|^2 = |W(z)|^2$$

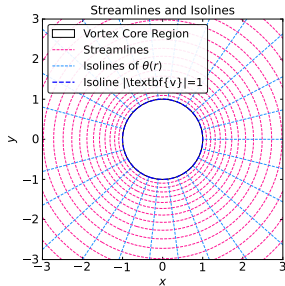
Conformal Mapping Strategy

Since the real and imaginary parts of any analytic complex function satisfy the Cauchy-Riemann equations, they automatically solve Laplace's equation. Thus, any analytic transformation $w = g(z)$ strictly preserves Laplace's equation and orthogonality of ψ and θ .

Methodological Strategy:

Solve potential flow around simple circular boundaries ($|z| = 1$) using classical electrodynamic techniques, then map to deformed physical defect geometries.





Isolated vortex ($n = +1$): circular streamlines and impenetrable normal region boundary at $r = 1$.

Physical Boundary Conditions

For an isolated, unpinned vortex at equilibrium:

- 1. Normal Region Boundary ($J_{\perp} = 0$):**
Supercurrent cannot penetrate the normal region ($\psi = 0$ on $|z| = 1$).
- 2. Critical Threshold Equality ($|\mathbf{v}| = 1$):**
The physical boundary of the normal region coincides precisely with the critical isoline.

Symmetry Preservation

Circular symmetry guarantees $|\mathbf{v}| = 1/r = 1$ exactly on $|z| = 1$. No external current $\implies$ exact self-consistency.

Superposition & Milne-Thomson Theorem

By leveraging the principle of linear superposition and the Milne-Thomson circle theorem, we can model circular defects. An external uniform current is represented by:

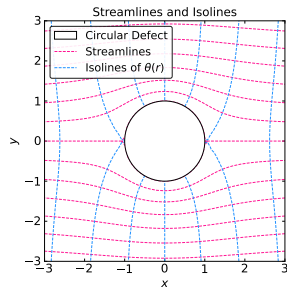
$$f_{\text{ext}}(z) = ij_0 z$$

Application to the Unit Defect

The function representing the current around the defect of radius $r_0 = 1$ reads:

$$f(z) = ij_0 \left(z + \frac{1}{z} \right)$$

On $|z| = 1$, the stream function $\psi = \text{Re}(f) = 0$. This mathematically enforces the physical boundary condition that superconducting Cooper pairs cannot flow into the normal region ($\mathbf{J}_\perp = 0$).



Uniform flow ($j_0 = 0.1$) deflected around a rigid circular obstacle.

Chapter 3

Single Vortex

Zeroth-order model & geometric feedback



Three Elementary Analytical Solutions

1. Isolated Vortex ($n = +1$):

$$f_v(z) = \log z, \quad W_v(z) = 1/z.$$

2. Uniform Current + Circular Hole:

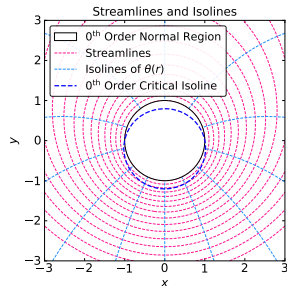
$$f_h(z) = ij_0(z + 1/z), \quad W_h(z) = ij_0(1 - 1/z^2).$$

3. Vortex Pinned in External Current:

$$f(z) = ij_0(z + 1/z) + \log z.$$

The Self-Consistency Challenge

When $j_0 \neq 0$, the critical isoline $|\mathbf{v}| = 1$ detaches from $|z| = 1$. Chapter 3 resolves this geometric mismatch via feedback.



Linear superposition: pinned vortex driven by uniform supercurrent ($j_0 = 0.1$). Critical isoline in red dashed.

Unperturbed Model ($r_0 = 1$, fixed at origin)

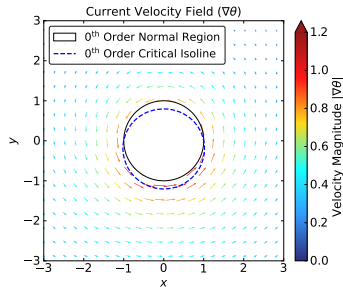
Baseline complex velocity around a rigid unit circle:

$$W(z) = ij_0 \left(1 - \frac{1}{z^2} \right) + \frac{1}{z}$$

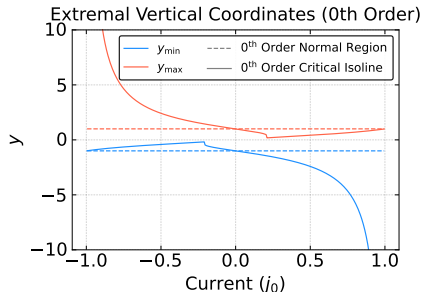
Constructive vs. Destructive Superposition

For $+j_0$ (rightward flow around counter-clockwise vortex):

- **South** ($y < 0$): External flow and circulation align $\Rightarrow |\mathbf{v}|$ is strongly amplified ($|\mathbf{v}| > 1$).
- **North** ($y > 0$): External flow and circulation oppose $\Rightarrow |\mathbf{v}|$ is suppressed ($|\mathbf{v}| < 1$).
- **Consequence:** The critical isoline $|\mathbf{v}| = 1$ deforms and shifts downward relative to the rigid circle.



Velocity gradient field ($j_0 = 0.1$): strong velocity enhancement at the southern edge, suppression at the north.



Extremal vertical coordinates ($y_{\max}$, $y_{\min}$) of the critical isoline vs. transport current j_0 .

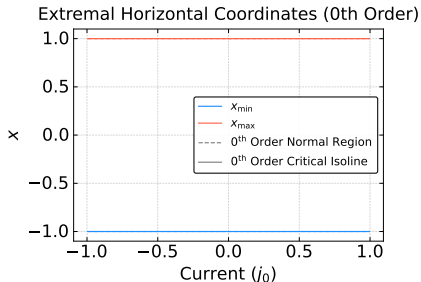
Non-Self-Consistent Boundary

Because $|\mathbf{v}| = 1$ (dashed red line) detaches from the unit circle (solid black), the model violates boundary self-consistency: the normal region ($\mathbf{J}_{\perp} = 0$) no longer matches the physical critical boundary ($|\mathbf{v}| = 1$).

Vertical Deformation Behavior

As transport current $|j_0|$ increases:

- **Upward Expansion ($y_{\max}$):** Grows linearly due to velocity suppression above the core.
- **Downward Shift ($y_{\min}$):** Plunges southward due to strong constructive enhancement.
- **Sharp Inflection:** A slope discontinuity appears at $|j_0| \approx 0.207$.



Horizontal coordinates across the core center ($x_{\max}, x_{\min}$) relative to shifted core center.

Horizontal Rigidity vs. Vertical Stretch

- While vertical stretching across the y-axis is severe, the horizontal diameter across the center ($x_{\max} - x_{\min}$) remains remarkably rigid.
- **Geometric Anisotropy:** The critical isoline does not expand uniformly as a circle, but stretches anisotropically.

Significance of the Kink at $|j_0| \approx 0.207$

A sharp kink appears simultaneously in both horizontal and vertical profiles near $|j_0| \approx 0.207$, indicating a topological transition in isoline curvature. Shifting a rigid circular core must account for this boundary distortion.

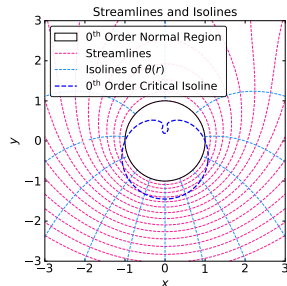
Analytic Evaluation Along $z = iy$

Complex velocity along imaginary axis ($z = iy$): $W(iy) = i(j_0 y^2 - y + j_0)/y^2 \Rightarrow |W(iy)|^2 = 1$.

Factoring as a difference of squares yields two quadratics:
 $[(j_0 - 1)y^2 - y + j_0] \cdot [(j_0 + 1)y^2 - y + j_0] = 0$.

Algebraic Root Regimes

- **4 Real Roots** ($|j_0| \leq \frac{-1+\sqrt{2}}{2} \approx 0.207$):
Both discriminants positive $\Rightarrow$ four intersections on y -axis (smoothly rounded top).
- **2 Real Roots** ($0.207 < |j_0| \leq 1.0$):
One discriminant negative $\Rightarrow$ two real roots remain, forming an inward pinching inflection (kink).



Inward pinching inflection: zoomed upper region ($y \approx 1$) showing the critical isoline curving inward toward the core as j_0 exceeds 0.207.

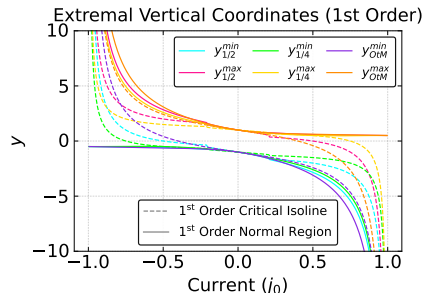
Adapted 1st-Order Potential

To mitigate non-self-consistency, we apply a spatial shift d to the rigid circular core along the y -axis, aligning it with the deformed critical isoline while keeping the vortex singularity fixed at $z = 0$:

$$f_{\text{new}}(z) = ij_0 \left(z + \frac{1}{z - d} \right) + \log z$$

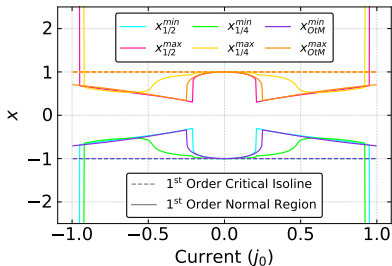
Three Shift Criteria Evaluated

- 1/2 Midpoint Model:** $d = \frac{y_{\text{max}} + y_{\text{min}}}{2}$
Centers the core at the exact midpoint of the critical isoline.
- 1/4 Conservative Model:** $d = \frac{y_{\text{max}} + y_{\text{min}}}{4}$
Conservative shift half-way toward the isoline midpoint.
- Extremal Model:** $d = y_{\text{min}} + 1$ ($j_0 \geq 0$)
Aligns lower core edge precisely with the bottom



Vertical coordinates post-feedback: dramatic realignment with the critical isoline boundary.

Extremal Horizontal Coordinates (1st Order)



Horizontal coordinates post-feedback relative to shifted core center across all models.

The Vertical-Horizontal Trade-Off

- Shifting the core vertically (y -axis) eliminates the massive 0th-order vertical mismatch ($y_{\min}$ and $y_{\max}$ alignment).
- However, because the true isoline stretches anisotropically, shifting a rigid circle slightly perturbs horizontal alignment across the core center ($x_{\max}, x_{\min}$).
- **Net Result:** 1st-order feedback provides an immense global improvement in boundary self-consistency across weak and moderate currents.

Exact Symmetry Under Reversal

The single-vortex configuration is invariant under simultaneous current reversal and spatial reflection across the horizontal axis:

$$j_0 \rightarrow -j_0 \iff y \rightarrow -y$$

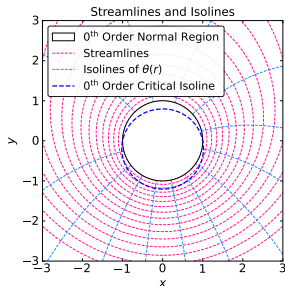
- Reversing the current mirrors the velocity field exactly across the x-axis.
- Consequently, the critical current magnitude required to destroy superconductivity is identical in both directions:

$$|j_{c,+}| = |j_{c,-}|$$

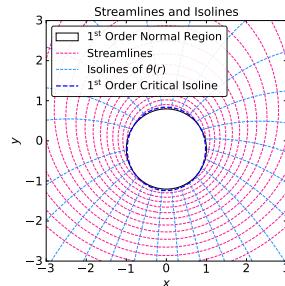
- $\implies$ **An isolated single vortex exhibits NO macroscopic non-reciprocity or diode effect.**

Low-Current Convergence ($|j_0| < 0.2$)

- As $j_0 \rightarrow 0$, all three correction models (1/2, 1/4, and Extremal) converge precisely toward exact self-consistency.
- In this weak-current regime, horizontal discrepancies are negligible and results are invariant to the specific correction model chosen.
- **Iterative Pathway:** This strong low-current convergence confirms that applying geometric feedback iteratively converges toward exact self-consistent boundaries.



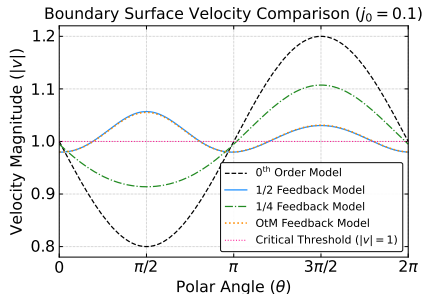
Single Vortex ($j_0 = 0.1$): Comparison of the complex potential streamlines and isolines around the defect for the 0th order model.



Single Vortex ($j_0 = 0.1$): Comparison of the complex potential streamlines and isolines for the 1/2 Feedback Model.

Weak Current Streamline Alignment

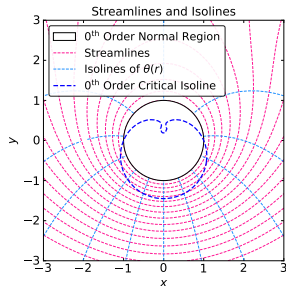
At $j_0 = 0.1$, the critical isoline remains nearly circular. The 1st-order shift completely eliminates the boundary mismatch.



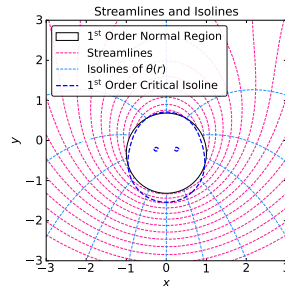
Single Vortex ($j_0 = 0.1$): Combined angular profile of the surface velocity $|\mathbf{v}(\theta)|$ evaluated along the boundary for all models.

Quantitative Boundary Assessment

- **0th-Order Violation:** Velocity deviates significantly ($|\mathbf{v}| \in [0.8, 1.2]$).
- **Optimal Realignment:** 1/2 and Extremal models dampen peaks, restoring $|\mathbf{v}| \approx 1$.
- **Model Comparison:** 1/2 and Extremal models are indistinguishable; 1/4 model is slightly less accurate but effective.



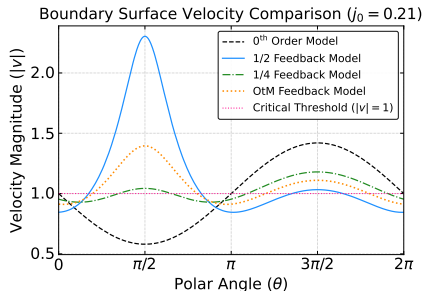
Single Vortex ($j_0 = 0.21$): Comparison of complex potential streamlines and isolines for the 0th order model.



Single Vortex ($j_0 = 0.21$): Comparison of complex potential streamlines and isolines for the 1/4 Feedback Model.

Onset of Topological Deformation

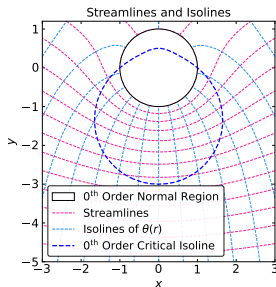
At $j_0 = 0.21$, crossing the algebraic root threshold (0.207) produces an inward curvature inflection at the top of the critical isoline. Shifting a rigid circle requires careful balancing to avoid over-displacing the core.



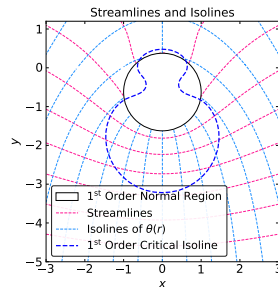
Single Vortex ($j_0 = 0.21$): Combined angular profile of the surface velocity $|\mathbf{v}(\theta)|$ evaluated along the boundary for all models.

Why Conservative 1/4 Feedback Excels Here

- **Rigid Circle Limitations:** A rigid circle cannot conform simultaneously to an inward-pinched top boundary and an expanded bottom boundary.
- **Over-Correction Risk:** Aggressive shifts (1/2 or Extremal) displace the core too far southward, exposing the northern boundary and inadvertently amplifying local velocity peaks.
- **Optimal Realignment:** The conservative 1/4 model achieves the best local boundary realignment, keeping $|\mathbf{v}|$ closest to unity.



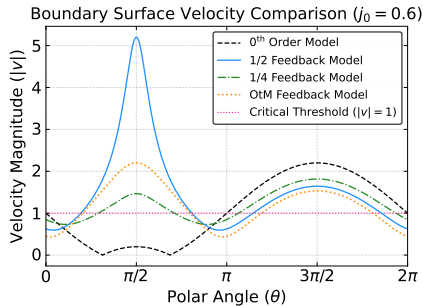
Single Vortex ($j_0 = 0.6$): Comparison of complex potential streamlines and isolines for the 0th order model.



Single Vortex ($j_0 = 0.6$): Comparison of complex potential streamlines and isolines for the 1/4 Feedback Model.

High-Current Regime Topology

Deep inside the high-current regime ($j_0 = 0.6$), the external transport field heavily dominates intrinsic circulation, forcing massive horizontal stretching across the critical velocity boundary.



Single Vortex ($j_0 = 0.6$): Combined angular profile of the surface velocity $|\mathbf{v}(\theta)|$. The large amplitude variations indicate massive boundary mismatch.

Fundamental Limitation of Rigid Spatial Shifts

- **Severe Non-Linearity:** The 1st-order rigid circular approximation breaks down because shifting a circle along y cannot absorb horizontal stretching ($x_{\max} - x_{\min}$).
- **Boundary Violation:** Velocity vectors cross the assumed circular perimeter at large angles across all feedback models.
- **Required Next Steps:** Achieving self-consistency at high currents requires dynamic core expansion $r_0(j_0)$ and higher-order conformal mappings ($w = g(z)$).

Chapter 4

Dipolar Configuration

Vortex-antivortex pair & Diode Effect



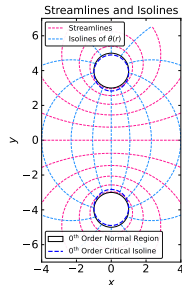
Dipolar Complex Potential (0^{th} Order)

Bound vortex (+1 at $+id/2$) and antivortex (-1 at $-id/2$) separated by distance d in uniform flow j_0 :

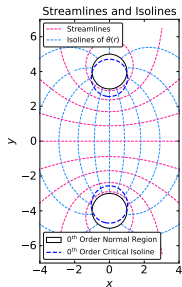
$$f(z) = ij_0 \left(z + \frac{1}{z - i\frac{d}{2}} + \frac{1}{z + i\frac{d}{2}} \right) + \log \left(z - i\frac{d}{2} \right) - \log \left(z + i\frac{d}{2} \right)$$

The Central Superconducting Channel

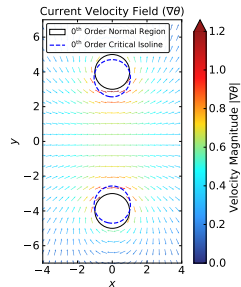
- A dissipationless channel separates the two cores across $y = 0$.
- If critical isolines touch across $y = 0$, the channel collapse $\Rightarrow$ vortex-antivortex annihilation into a resistive state.
- Linear superposition is exact when $d \gg r_0 = 1$.



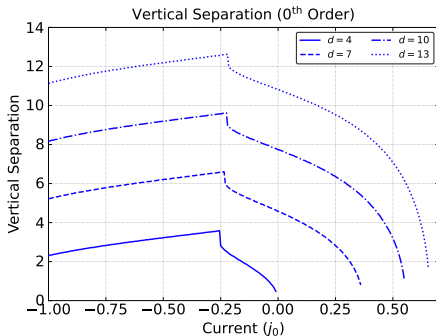
Dipolar Configuration ($j_0 = 0, d = 8$): Asymmetric current in a vortex-antivortex pair separated by an initial center-to-center distance $d = 8$ without external current.



Streamlines and isolines.



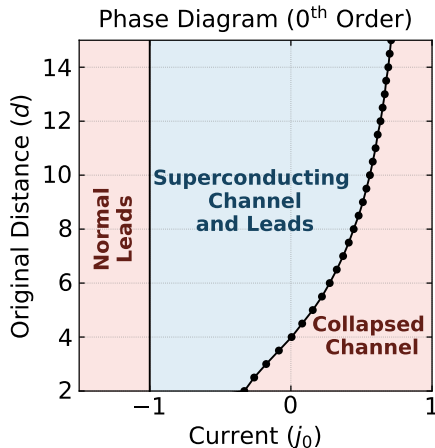
Gradient vector field.



Dipolar Configuration (0th Order): Vertical channel gap between disjoint isolines vs. j_0 across initial separations d .

Key Physical Observations

- Larger d : weaker mutual interaction $\implies$ channel persists to higher currents.
- $j_0 > 0$: gap decreases rapidly (constructive amplification $\rightarrow$ early collapse).
- $j_0 < 0$: gap decreases very slowly (destructive suppression $\rightarrow$ open channel).
- Discontinuity at $|j_0| \approx 0.207$: marks the algebraic topological kink.



Phase diagram mapping transition boundaries between separated and merged critical isolines.

Stability Thresholds

- **Critical Distance ($d_{\text{crit}}^{(0)} \approx 4$):**
Minimum separation for disjoint lobes at $j_0 = 0$.
- **Dynamic Stabilization:**
For $d < 4$, applying a negative current ($j_0 < 0$) can destructively suppress the velocity field sufficiently to re-separate the merged lobes into distinct topological domains.

Adapted 1st-Order Dipolar Potential

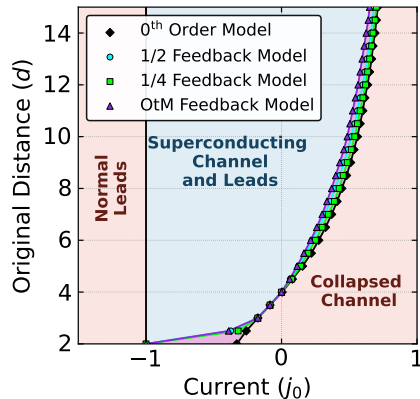
normal regions shift symmetrically to $\pm i \frac{d_{\text{new}}}{2}$, while topological singularities remain fixed at $\pm i \frac{d}{2}$:

$$f_{\text{new}} = ij_0 \left(z + \frac{1}{z - i \frac{d_{\text{new}}}{2}} + \frac{1}{z + i \frac{d_{\text{new}}}{2}} \right) + \log \frac{z - i \frac{d}{2}}{z + i \frac{d}{2}}$$

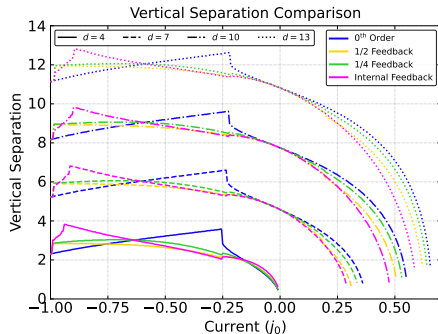
Internal vs. Midpoint Feedback Models

- **Internal Model:** Shifting based strictly on inner extremal coordinate facing the central channel.
- **Midpoint/Conservative Models:** Shifts cores inward when $j_0 > 0$, narrowing the gap and accelerating collapse vs. baseline ($\Rightarrow$ lower i_{c1})

Phase Diagram (1st Order)



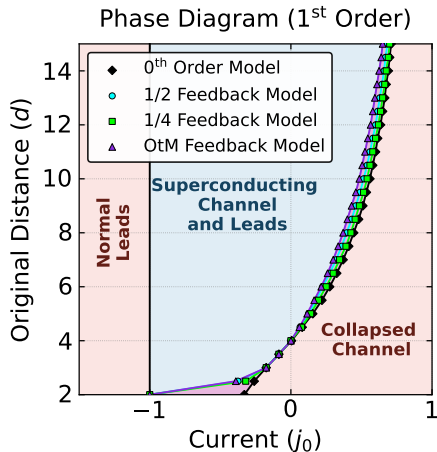
Unified 1st-Order Phase Diagrams: all models agree qualitatively on directional asymmetry.



Vertical separation vs j_0 comparing baseline to geometric feedback models.

Channel Gap Dynamics

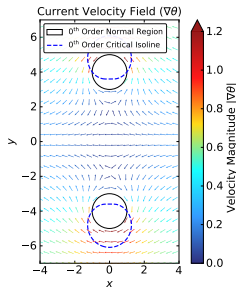
- Feedback physically shifts normal regions inward when $j_0 > 0$, resulting in a smaller channel gap compared to the 0th-order baseline.
- **Model Hierarchy:** Internal feedback dictates the most aggressive shift, triggering the earliest collapse. 1/4 feedback is the most conservative.



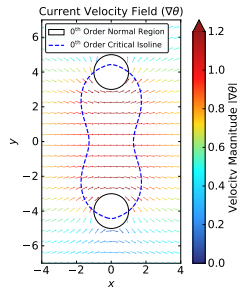
Phase diagrams comparing transition thresholds of all feedback models.

Robustness of Phase Boundaries

- The system is significantly more susceptible to channel collapse than predicted by the rigid baseline.
- **Directional Asymmetry:** The macroscopic critical threshold depends strictly on the current sign.



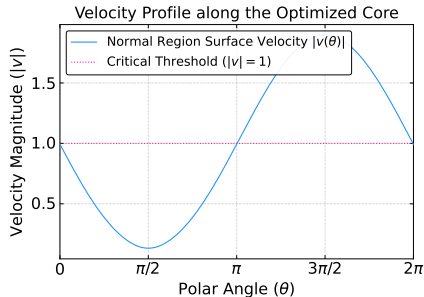
Superconducting Diode Effect ($d = 8$): Comparison of the current gradient vector fields under external current $j_1 = -0.5$ in the 0th order model.



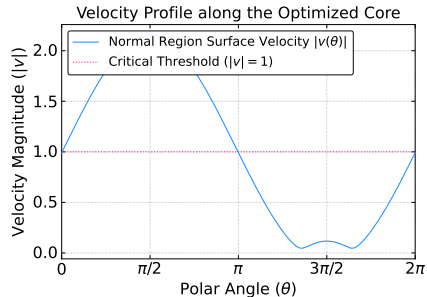
Superconducting Diode Effect ($d = 8$): Comparison of the current gradient vector fields under external current $j_2 = 0.5$ in the 0th order model.

Breaking of Spatial Inversion Symmetry

Because the dipole breaks vertical symmetry along y , current direction strictly controls interference in the central channel. Transport is **strongly non-reciprocal**: supercurrent flows dissipationlessly in one direction but collapses the gap in the other.



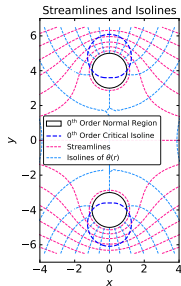
Superconducting Diode Effect ($d = 8$): Comparison of the surface velocity along the boundary of the lower normal region under external current $j_1 = -0.5$.



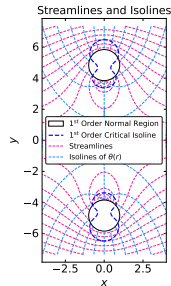
Superconducting Diode Effect ($d = 8$): Comparison of the surface velocity along the boundary of the lower normal region under external current $j_2 = 0.5$.

Direction-Dependent Critical Threshold ($j_{c,+} \ll |j_{c,-}|$)

For any transport current $j_0 \in (j_{c,+}, |j_{c,-}|)$, forward flow ($+j_0$) collapses the channel into a resistive state while reverse flow ($-j_0$) conducts dissipationlessly. Pinned topological defects inherently induce a Superconducting Diode Effect.



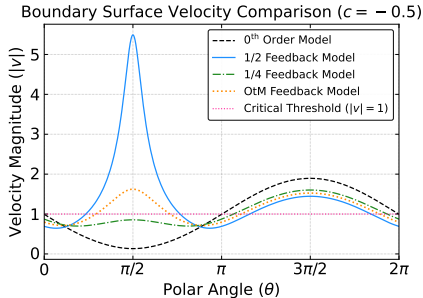
*Dipolar Configuration ($j_0 = -0.5, d = 8$):
 Application of the 0th Order model under a strong
 negative transport current.*



*Dipolar Configuration ($j_0 = -0.5, d = 8$):
 Application of the 1/2 Feedback model under a
 strong negative transport current.*

Strong Negative Current Regime

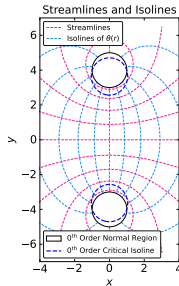
Because the critical isolines remain topologically disjoint under reverse flow ($j_0 = -0.5$), unlike the macroscopic merging at $j_0 = +0.5$ —the independent two-circle assumption remains valid, allowing geometric feedback to operate safely.



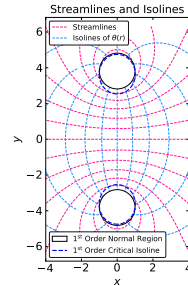
Dipolar Configuration ($j_0 = -0.5, d = 8$): Combined angular profile of the surface velocity $|\mathbf{v}(\theta)|$ under strong negative external current.

Quantitative Boundary Assessment

- **0th-Order Violation:** Unperturbed baseline exhibits substantial velocity fluctuations around the circular boundary due to strong external flow ($|j_0| = 0.5$).
- **Feedback Correction:** Implementing 1st-order spatial shifts (1/4 and Internal) measurably improves boundary self-consistency toward $|\mathbf{v}| \approx 1$.
- **Strong-Drive Limit:** While measurable improvements occur, $|j_0| = 0.5$ represents a strong drive where residual fluctuations persist.



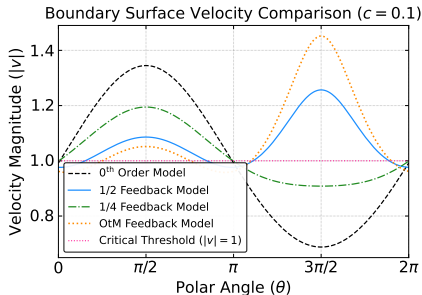
Dipolar Configuration ($j_0 = 0.1, d = 8$): Application of the 0th Order baseline model.



Dipolar Configuration ($j_0 = 0.1, d = 8$): Application of the 1/4 Feedback model, demonstrating the spatial shift of the physical normal regions.

Weak Forward Current Regime

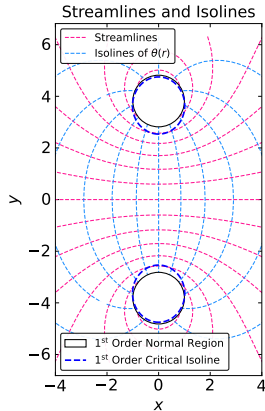
At weak positive current ($j_0 = 0.1$), the critical isolines are widely separated across the central channel ($d = 8$). All three geometric feedback models prove highly effective at mitigating boundary non-self-consistency.



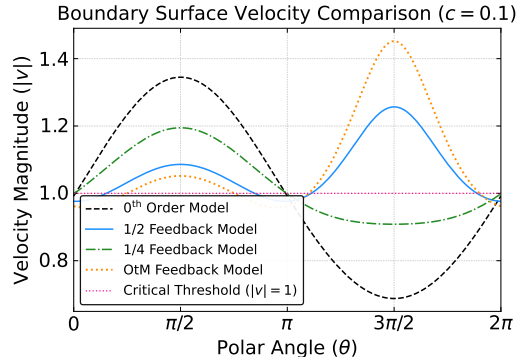
Dipolar Configuration ($j_0 = 0.1, d = 8$): Combined angular profile of the surface velocity $|\mathbf{v}(\theta)|$ evaluating the boundary alignment for all models.

Optimal Low-Current Realignment

- **Remarkable Convergence:** All feedback methods (1/2, 1/4, Internal) eliminate 0th-order velocity variations, restoring $|\mathbf{v}| \approx 1$ uniformly.
- **Subtle Distinction:** The Internal model shows slightly larger residual velocity fluctuations along the core perimeter compared to 1/2 and 1/4.
- **Global Agreement:** Macroscopic streamlines and supercurrent deflections across the channel are indistinguishable among all three models.

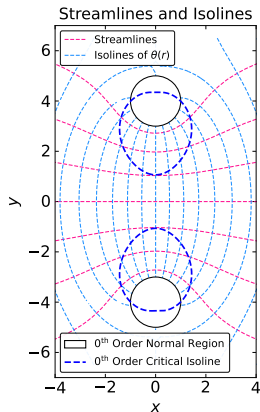


1/4 Feedback Model.

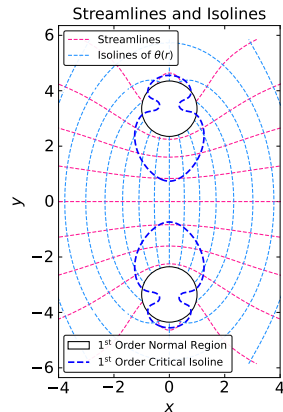


Surface velocity $|\mathbf{v}(\theta)|$ comparison.

Dipolar Configuration ($j_0 = 0.1, d = 8$): Application of the feedback models compared to the unperturbed baseline, demonstrating the spatial shift of the physical normal regions.

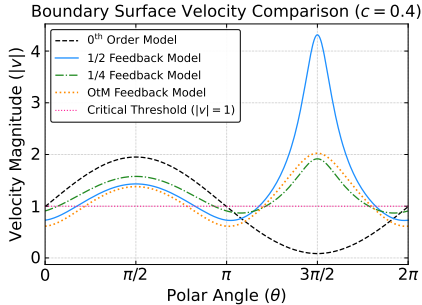


0th Order Model.



1/4 Feedback Model.

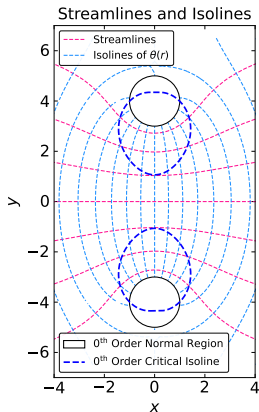
Dipolar Configuration ($j_0 = 0.4, d = 8$): Application of the feedback models compared to the unperturbed baseline, demonstrating the spatial shift of the physical normal regions.



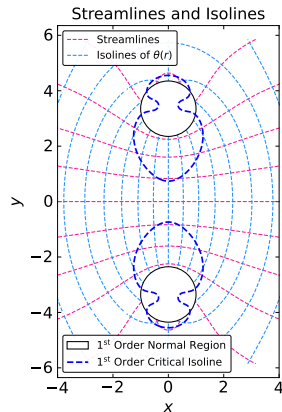
Dipolar Configuration ($j_0 = 0.4, d = 8$): Combined angular profile of the surface velocity $|\mathbf{v}(\theta)|$ under high external current.

Why Aggressive Shifts Fail Near Threshold

- **Premature Collision:** Shifting cores inward via aggressive models (1/2 or Internal) forces physical circles to collide prematurely across $y = 0$, causing unphysical channel collapse.
- **Conservative Superiority:** The 1/4 model performs best because its conservative displacement preserves open channel topology while mitigating boundary velocity peaks.
- **Two-Circle Limit:** Demonstrates that rigid circle shifts must give way to dynamic shape deformations when isolines stretch near critical thresholds.



0th Order Model.



1/4 Feedback Model.

Dipolar Configuration ($j_0 = 0.4, d = 8$): The 1/4 Model preserves disjoint topology, preventing premature channel collapse seen in the baseline model.

Methodological & Single-Vortex Synthesis

1. Conformal Mapping for Superfluids:

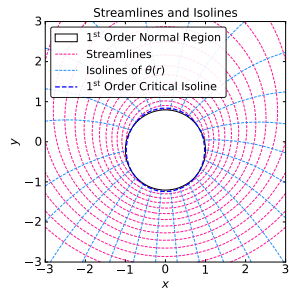
Because macroscopic phase obeys Laplace's equation ($\nabla^2\theta = 0$), conformal mapping provides an exact analytical framework for modeling 2D superfluid velocity around boundaries.

2. Boundary Realignment:

First-order geometric feedback successfully realigns normal regions with deformed critical isolines across weakly driven regimes ($|j_0| < 0.2$).

3. Strict Single-Vortex Reciprocity:

Isolated vortices obey exact current reversal symmetry ($j_0 \rightarrow -j_0 \Leftrightarrow y \rightarrow -y$), yielding strictly reciprocal transport ($|j_{c,+}| = |j_{c,-}|$).



Single vortex feedback ($j_0 = 0.1$): exact boundary self-consistency and spatial reflection symmetry.

Conclusions: Dipolar Non-Reciprocity & Diode Effect

Dipolar Diode Synthesis

1. Broken Spatial Symmetry:

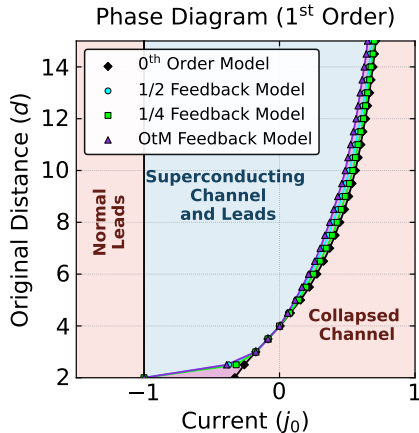
Interacting vortex-antivortex pairs break spatial inversion along y . Constructive vs. destructive interference strictly dictates directional channel stability.

2. Superconducting Diode Effect:

Directional channel collapse establishes a robust critical threshold gap ($j_{c,+} \ll |j_{c,-}|$), enabling dissipationless forward blocking and reverse conduction.

3. Robustness Across Models:

While specific shift criteria (1/2, 1/4, Internal) vary near threshold, all models agree on the phase diagram structure and directional diode property.



Unified phase boundaries: directional non-reciprocity is a robust physical property across all approximations.

Methodological Refinements

- **Iterative Self-Consistent Loop:** Transition from single-step feedback to automated iterations. For $|j_0| < 0.2$, this converges mathematically to exact self-consistent boundaries.
- **Dynamic Core Expansion** $r_0(j_0)$: Overcome rigid-circle limits by allowing effective core radius to expand/shrink with local supercurrent density.
- **Higher-Order Conformal Mappings:** Map the unit circle onto elliptical profiles to exactly match high-current ($|j_0| \geq 0.4$) isoline stretching.

Experimental Transport Devices

- **Bridging the Gap:** Capturing exact morphological deformations bridges the gap between idealized potential flow theory and the pronounced non-reciprocity of the superconducting diode effect.
- **Quantitative Comparisons:** This advancement will allow for direct, quantitative comparisons with modern experimental transport devices.

Analytical + Numerical Implementation

Developed and validated entirely in Wolfram Mathematica

Thank You

Questions & Discussion

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